OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester I Examinations 2022–2023

Course Instance Codes(s)	4BMS2, 4BS2, 4BPT2, 1HA1, 1AL1
Exam(s)	4th Science.
Module Codes	MA4102
Module	**Algebraic Foundations of Quantum Computing**
Paper No	1
External Examiner	Prof. Colva Roney-Dougal
Internal Examiner(s)	Prof. Götz Pfeiffer
	⋆ Dr Mark Howard
Instructions:	**Answer FOUR questions**
Duration	2 Hours
Number of Pages	3 (including this page)
Discipline	Mathematics

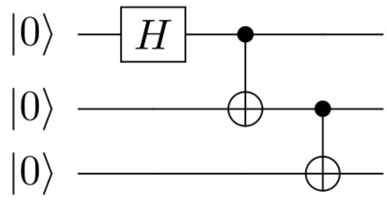
Requirements:

Release in Exam Venue	Yes ☑	No ☐
Release to Library	Yes ☑	No ☐
Statistical/ Log Tables:	State Examination Commission's *Formulae and Tables*	
Handout	None	
Cambridge Tables	None	
Graph Paper	None	
Log Graph Paper	None	
Other Materials	None	
Graphic material in colour	Yes ☐	No ☑

1. (a) [8 Marks] Write out the truth tables for each of $CNOT_{1,2}$ and $CNOT_{2,1}$. For example, the first line of each truth table would be $|0, 0\rangle \mapsto |0, 0\rangle$.

   (b) [9 Marks] Show that surrounding a CNOT gate with Hadamard gates switches the role of control and target qubit, e.g., $(H \otimes H)CNOT_{1,2}(H \otimes H) = CNOT_{2,1}$.

   (c) [8 Marks] What is the output state for the following circuit



2. Consider the qutrit state given by

$$|\psi\rangle = \frac{1}{3}(2\,|0\rangle + i\,|1\rangle + 2i\,|2\rangle)$$

   (a) [6 Marks] If a measurement is made in the computational basis $\{|0\rangle, |1\rangle, |2\rangle\}$, what is the probability of seeing the outcome "2", and what is the post-measurement state in that case?

   (b) [6 Marks] Write out the density matrix $\rho = |\psi\rangle\langle\psi|$. What are the eigenvalues of ρ?

   (c) [6 Marks] What is the von Neumann entropy S of $\rho = |\psi\rangle\langle\psi|$? Write down a state ρ' that has different von Neumann entropy than ρ and give the value of $S(\rho')$.

   (d) [7 Marks] If we apply to $|\psi\rangle$ the unitary

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}$$

   what is the subsequent probability of seeing the outcome "2" in a computational basis measurement?

3. (a) [13 Marks] Let $|a\rangle$ be an n-qubit basis state (i.e., $a \in \{0,1\}^n$). Define $|\phi\rangle$ to be

$$|\phi\rangle = \frac{1}{\sqrt{2^n - 1}} \sum_{x \in \{0,1\}^n, x \neq a} |x\rangle .$$

   In the 2D subspace spanned by $\{|a\rangle, |\phi\rangle\}$, we may write the Grover iterate as a 2×2 matrix G. For searching a list of length four, write out the matrix G, the superposition of all basis states $|\psi\rangle$, and finally $G|\psi\rangle$ (all in the $\{|a\rangle, |\phi\rangle\}$ basis).

   (b) [12 Marks] Calculate the Quantum Fourier Transform of the following qudit states

$$(i)\quad \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ i \end{pmatrix} \qquad\qquad (ii)\quad \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ e^{\frac{2\pi i}{3}} \\ e^{\frac{2\pi i}{3}} \end{pmatrix}$$

4. (a) [8 Marks] Prove that the state $|\beta_{0,0}\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is entangled.

   (b) [9 Marks] Prove the quantum no-cloning theorem i.e., the fact there does not exist a 2-qubit unitary that maps

$$|\psi\rangle\,|0\rangle \mapsto |\psi\rangle\,|\psi\rangle$$

   for every qubit state $|\psi\rangle$.

   (c) [8 Marks] Show that unitaries cannot delete information i.e., there does not exist a single-qubit unitary that maps

$$|\psi\rangle \mapsto |0\rangle$$

   for every qubit state $|\psi\rangle$.

5. Let $U_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $|\psi\rangle = U_\theta\,|0\rangle$ and $|\psi^\perp\rangle = U_\theta\,|1\rangle$

   (a) [8 Marks] Show that $|\beta_{0,0}\rangle = \frac{1}{\sqrt{2}}(|\psi,\psi\rangle + |\psi^\perp,\psi^\perp\rangle)$ where as usual $|\beta_{0,0}\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

   (b) [9 Marks] Suppose Alice and Bob start with the state $|\beta_{0,0}\rangle$. Alice applies U_θ^{-1} on her qubit and then measures in the computational basis. What pure state does Bob have if her measurement is zero? What pure state does Bob have if her measurement is one?

   (c) [8 Marks] Suppose that something called a "parity measurement" on two qubits comprises two projectors $\Pi_{\text{even}} = |00\rangle\langle 00| + |11\rangle\langle 11|$ and $\Pi_{\text{odd}} = |01\rangle\langle 01| + |10\rangle\langle 10|$. Show that the parity measurement obeys the requirements of a projective measurement, and calculate the probability of getting the outcome "even" if the state $|\beta_{0,0}\rangle$ is measured.